Econometrics I
PPS-04
Multiple Choice Questions
(Chapters 7-8-9)
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1. **Suppose that a model for city crime rates is formulated as follows:**

$$crimerate = \beta_0 + \delta_0 West + \gamma_0 North + \alpha_0 South + \beta_1 unem + u$$

where unem is the unemployment rate in the city, and West, North, South are dummy variables indicating the geographic locations of the city. Each city is classified according to its geographical locations as east, west, north and south. What is the base group?

A. East B. West C. North D. South E. We don't know.

2. **Consider the following estimation results in equation form:**

$$\widehat{\text{sleephours}} = 8.51 + 0.21 \text{ male} - 0.17 \text{ workhours}$$

$$n = 706, \qquad R^2 = 0.12,$$

where sleephours is the average length of sleep per day (in hours), workhours is the average length of work per day (in hours), and male is a dummy variable that equals 1 if the individual is male and 0 otherwise. According to these results we can say

A. If workhours increases 2 hours then sleep is predicted to decrease by 0.34 hours.
B. Men sleep approximately 0.21 hours more than women, ceteris paribus.
C. Women's intercept is 8.51.
D. Men's intercept is 8.72.
E. All of the above.

3. **In the following model score represents grades from Econometrics I (out of 100 points). We group students into three categories according to their class attendance. Each student belongs to one of these groups: D_h: high attendance, D_m: medium attendance, D_l: low attendance. When we set the low attendance as the base group we obtain:**

$$\widehat{\text{score}} = 30 + 20D_m + 30D_h$$

If we set the high attendance, D_h, as the base group, which of the following regression do we obtain?

A.
$$\widehat{\text{score}} = 30 + 30D_l + 20D_m$$

B.
$$\widehat{\text{score}} = 50 - 30D_l - 20D_m$$

C.
$$\widehat{\text{score}} = 60 + 20D_l + 10D_m$$

D.
$$\widehat{\text{score}} = 60 - 10D_l - 20D_m$$

E.
$$\widehat{\text{score}} = 60 - 30D_l - 10D_m$$

4. Suppose that the consumption of an individual in Eyeland depends on the eye color and some other individual-specific factors which can be measured in a quantitative manner. If there are five different eye colors in Eyeland, **how many dummy variables for eye color should be added to a regression with a constant term for consumption determination in Eyeland?**

A. 2

B. 3

C. 5

D. 4

E. 6

5. **Consider the following regression model**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

where y is a binary variable. The slope parameter on x_1, that is, β_1 measures

A. the predicted change in the value of y when x_1 changes by one unit, ceteris paribus.

B. the predicted change in the probability of success when x_1 changes by one unit, ceteris paribus.

C. the predicted change in the probability of failure when x_1 changes by one unit, ceteris paribus.

D. the predicted change in the probability of failure when x_1 changes by 100 per cent, ceteris paribus.

E. the predicted change in the value of y when x_1 changes by one per cent, ceteris paribus.

6. **Consider the following regression model:**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

In which of the following cases, is y a <u>discrete</u> variable?

A. y indicates the growth rate of the gross domestic product

B. y indicates logarithmic price of smart phones

C. y indicates total rainfall in a given year in a country

D. y indicates the number of children in a family

E. y indicates monthly water consumption in a city

7. **Consider the model**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_4 + \beta_4 x_4 + u,$$

$$\mathbf{E}[u|x_1, x_2, x_3, x_4] = 0,$$
$$\mathbf{Var}[u|x_1, x_2, x_3, x_4] = \sigma^2 x_2^2.$$

Which of the following is <u>correct</u>?

A. OLS is efficient

B. WLS is more efficient than OLS

C. The model cannot be estimated by OLS

D. OLS estimator is BLUE.

E. The model is homoskedastic

8. **Consider the following regression model:**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + u$$

We have a sample of size $n = 90$. Which of the following is the correct distribution of the RESET test statistic?

A. $F(2,\ 72)$

B. $F(2,\ 85)$

C. χ^2_4

D. χ^2_5

E. $F(2,\ 83)$

9. **Consider the results of the following regression model:**

$$\widehat{\mathrm{y}} = \underset{(09.85)}{1.28} + \underset{(0.16)}{0.48}\,\mathrm{x}_1 - \underset{(0.15)}{0.85}\,\mathrm{x}_2 + \underset{(0.92)}{0.45}\,\mathrm{x}_3$$
$$n = 200 \quad R^2 = 0.42$$

Using the residuals $\widehat{\mathrm{u}}$ from the regression above, we obtained:

$$\widehat{\mathrm{u}}^2 = \underset{(0.002)}{0.0002} + \underset{(0.03)}{0.03}\,\mathrm{x}_1 + \underset{(0.10)}{0.08}\,\mathrm{x}_2 + \underset{(0.18)}{0.04}\,\mathrm{x}_3$$
$$n = 200 \quad R^2 = 0.048$$

We want to compute the BP heteroskedasticity test. Which of the following is <u>correct</u>?

A. BP heteroskedasticity test statistic is 9.6 and follows χ^2_{200} distribution.

B. BP heteroskedasticity test statistic is 96 and follows χ^2_{200} distribution.

C. BP heteroskedasticity test statistic is 48 and follows F distribution.

D. BP heteroskedasticity test statistic is 9.6 and follows χ^2_3 distribution.

E. BP heteroskedasticity test statistic is 4.8 and follows χ^2_3 distribution.

10. **Consider the results of the following regression model:**

$$\widehat{\mathrm{y}} = \underset{(09.85)}{1.28} + \underset{(0.16)}{0.48}\,\mathrm{x}_1 - \underset{(0.15)}{0.85}\,\mathrm{x}_2 + \underset{(0.92)}{0.45}\,\mathrm{x}_3$$
$$n = 200 \quad R^2 = 0.42$$
$$RESET = 1.85 \quad (pvalue = 0.13)$$

where the p-value of the RESET test is provided in the parenthesis next to it. Based on this, what is your decision? (use 5% significance level)

A. The functional form is correctly specified.

B. The model is homoscedastic.

C. The error terms are not normally distributed.

D. The functional form is misspecified.

E. The model is heteroscedastic.

11. **Consider the model**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

Let the residuals and fitted values from that regression be $\hat{u}$ and $\hat{y}$, respectively. Which one of the following models is the correct test regression for the BP heteroskedasticity test?

A.
$$\hat{u}^2 = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + error$$

B.
$$\hat{u}^2 = \alpha_0 + \alpha_1 \hat{y} + \alpha_2 \hat{y}^2 + + error$$

C.
$$\hat{u}^2 = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_1^2 + \alpha_4 x_2^2 + error$$

D.
$$\hat{u}^2 = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_1^2 + \alpha_4 x_2^2 + \alpha_5 x_1 x_2 + error$$

E.
$$\hat{u} = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + error$$

12. **In the model**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u,$$

it is known that x_2 and x_3 are related to the error variance as follows:

$$Var(u|x_1, x_2, x_3) = \sigma^2 x_2^2 x_3.$$

We know that, in this case, OLS estimators are ..... and the ..... transformation has a constant variance. In order to obtain the transformed variables, we have to multiply each variable by ..... .
Fill in the blanks.

A. efficient, GLS-WLS, $x_1\sqrt{x_3}$

B. efficient, FGLS, $x_1\sqrt{x_3}$

C. inefficient, GLS-WLS, $\frac{1}{x_1\sqrt{x_3}}$

D. inefficient, GLS-WLS, $\frac{1}{x_2\sqrt{x_3}}$

E. inconsistent but efficient, FGLS, $\frac{1}{x_2\sqrt{x_3}}$

13. **Fill in the blanks: Measurement errors in the dependent variable does not lead to ..... of the OLS estimator as long as the measurement error is ..... with the independent variables. But it may lead to .....**

A. inconsistency, uncorrelated, functional form misspecification

B. inconsistency, correlated, lower variance.

C. inconsistency, uncorrelated, higher variance.

D. inefficiency, uncorrelated, functional form misspecification.

E. inefficiency, correlated, higher variance.

14. **Which of the following may not be classified as a data problem?**

A. Outliers

B. Functional form misspecification

C. Measurement errors

D. Nonrandom samples

E. Missing data